Pricing the Impermanent:
An Option-Implied Framework for Hedging Uniswap v3 Liquidity-Provider Loss

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Abstract. Topaz Blue and Bancor (2021) showed that, in aggregate, Uniswap v3 liquidity providers lost more to impermanent loss (IL) than they earned in fees — an empirical, backward-looking diagnosis. We extend it 3 ways:

- An independent **position-level re-measurement** across 5 major ETH pools confirms the result — **$50.7m of IL across 27,529 closed positions, every pool negative** — using a cleaner estimator built on NonfungiblePositionManager token-IDs, valued at each position's exact exit price.
- We formalise IL as a **short-gamma options position** and price the protection off the live Deribit volatility surface via SABR-calibrated static replication, treating the capped 'Limited' product as a **first-passage barrier** rather than a European condor.
- We quantify the rebalancing **'gamma bleed'** the original authors flagged as future work, with a Monte-Carlo calibrated to realised volatility, fee APR, and TVL computed live on **Dune Analytics**.

Why it matters. Because IL is a priced, tradeable **short-gamma** exposure, an LP can in principle **finance the protection out of fee income** — converting a structural, variable loss into a fixed, hedgeable cost. A second result, directly actionable for product design, is that the band-edge cap is **asymmetric**: a linearly symmetric 70–130% range locks a *deeper* loss on the downside (–10.7%) than the upside (–6.1%). The result is a forward-looking, market-implied framework for pricing and hedging LP loss.

1. The unhedged LP loses — re-measured at the position level

The Topaz Blue cohort (17 pools, 43% of TVL) earned $199.3m in fees but suffered $260.1m of IL, a net $60.8m below HODL, leaving 49.5% of LPs with negative returns. We reproduce the direction of that finding independently and at the position level. Reconstructing every position's deposits and withdrawals from NonfungiblePositionManager *IncreaseLiquidity*/*DecreaseLiquidity* events keyed by token-ID, restricting to single deposit–withdraw cycles held at least 1 day (excluding intra-block 'flash' LPs, the only group the original study found did not lose), and valuing each position's terminal basket at its exact exit-minute price, we measure the realised IL of 27,529 closed positions across 5 flagship ETH pools. Every pool is negative; the total is –$50.7m (Fig1). **This section measures IL in isolation** — fee income is introduced separately in Section 3, where the net LP outcome is evaluated; 'every pool lost IL' is not the same claim as 'every LP lost money net of fees'.

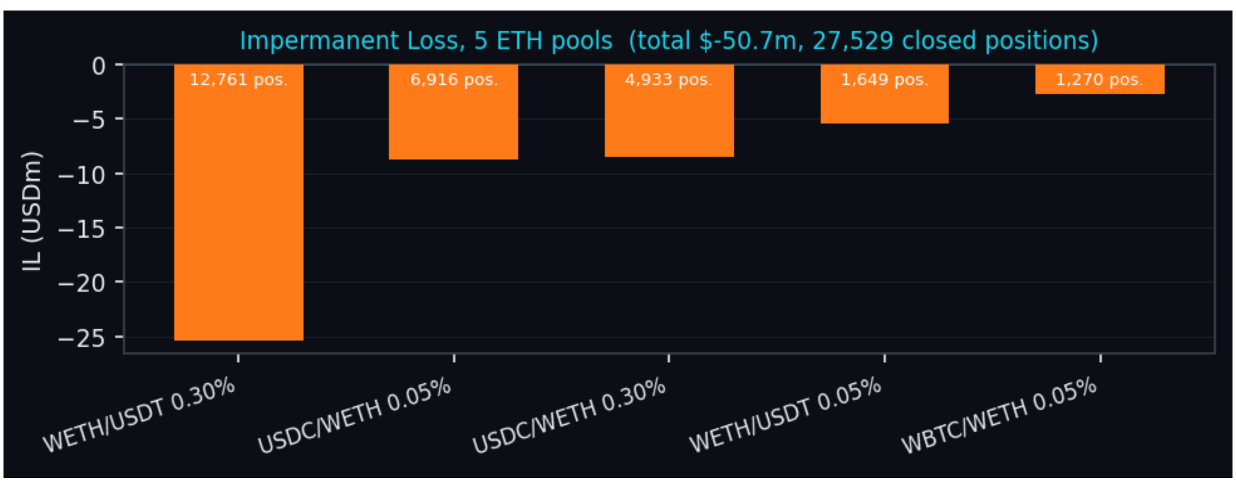

Fig1. Position-level measured IL by pool (Dune; trailing year). Counts above each bar are closed positions.

2 estimator points are new relative to the original work. (i) **Pool-level Mint/Burn events cannot isolate positions**: the position owner is the NFT manager contract, collapsing thousands of users at 1 tick range into 1 blob; token-ID reconstruction is required. (ii) Because *IL = (W – D)·price* is **linear in price**, it equals the true loss only **at the exit price**; valuing at a daily average flips signs on high-churn pools. Exit-minute pricing resolved the last positive outlier and yielded uniformly negative IL.

Pool	Closed pos.	Measured IL ($m)	Avg IL / pos. ($)	% of total IL
WETH/USDT 0.30%	12,761	-25.4	1,987	50%
USDC/WETH 0.05%	6,916	-8.8	1,270	17%
USDC/WETH 0.30%	4,933	-8.5	1,731	17%
WETH/USDT 0.05%	1,649	-5.4	3,263	11%
WBTC/WETH 0.05%	1,270	-2.7	2,102	5%
All 5 pools	**27,529**	**-50.7**	**1,842**	**100%**

Table 1. Per-pool breakdown. The loss is **not concentrated in one pool**: per-position IL is comparable across pools ($1.3k–$3.3k), and although WETH/USDT 0.30% is ~half the dollar total, that reflects its far larger *position count*, not outsized individual positions. A holding-behaviour check (Q7) reinforces this: by pair, the median closed position is $2,598–$5,926 (retail-scale, not whales) and the average holding period is ~18 days (multi-week, not flash). We do not report a top-tail statistic, so a few large losers *within* a pool cannot be fully ruled out; the comparable per-position and median figures are consistent with a wide base rather than a few whales. Pool-wide annual fees (Q2/Q4) are used in Section 3, not netted here, to avoid conflating IL with net P&L.

2. IL is short gamma: pricing the hedge off the smile

An LP is short convexity: the pool sells the appreciating asset into a rally, ending with less of the winner — a concave payoff. The protection an LP would buy is the mirror image, a **long** options position. By the Carr–Madan identity, any twice-differentiable payoff is a static strip of options: $E^Q[g] = g(F) + \int g''(K)\cdot O(K)\, dK$, with O the out-of-the-money put (K<F) or call (K>F) — the undiscounted, forward-measure option value (we ignore discounting over short crypto tenors). For full-range (V2) IL the leading-order premium is the variance-swap value $\sigma^2 T/8$; for concentrated (V3R) IL the same strip is amplified inside the band by capital efficiency. We evaluate the integral under the SABR-fitted Deribit density (Hagan et al., 2002), matching indicative desk-style V2/V3R IL quotes to within a few basis points under comparable assumptions.

The capped ‘Limited’ (V3L) product is **not** a European iron condor: it **locks** the IL at the band edge on first touch, a path-dependent **barrier / double-no-touch** payoff priced by first-passage Monte-Carlo. A static European strip prices only the terminal-clamp approximation; the gap to the barrier value is the path-dependence a vanilla strip cannot replicate — a distinction absent from the original report. A second, previously unremarked property is the **asymmetry of the band-edge cap** (Fig2): IL is symmetric under the price ratio r ↔ 1/r, not under linear distance, so a linearly symmetric band such as 70–130% caps a *deeper* loss on the downside (–10.7%) than the upside (–6.1%); the caps match only for a reciprocal-symmetric band (e.g. 70–143%).

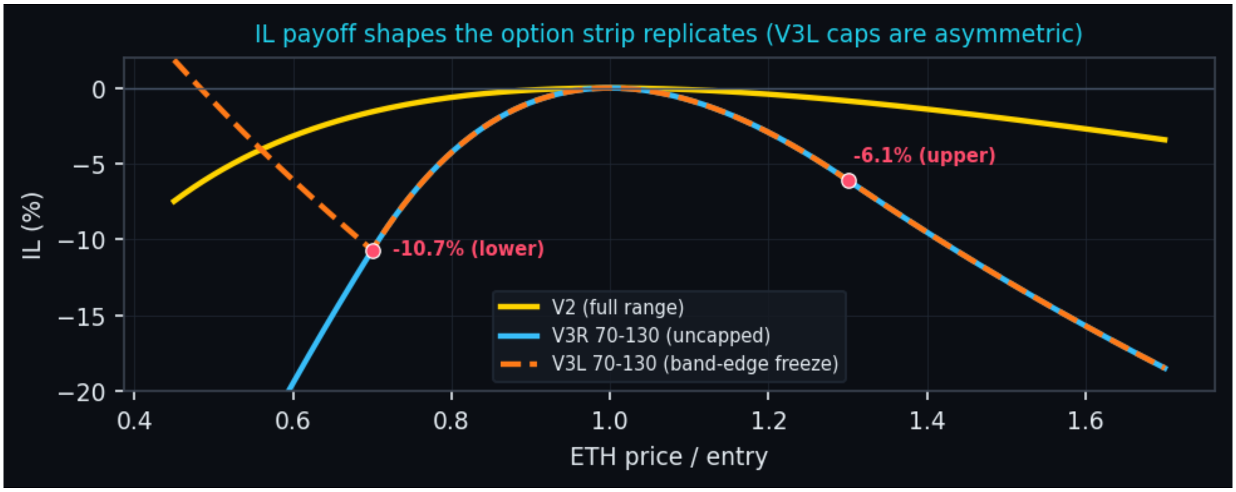

Fig2. IL payoff curves the option strip replicates; red points mark the asymmetric V3L first-touch caps.

Payoff glossary. V2 — full-range LP; IL concave and uncapped on both sides. **V3R** (‘Regular’) — liquidity concentrated in a band [a,b]; IL amplified in-band by capital efficiency, uncapped beyond it. **V3L** (‘Limited’) — identical in-band, but IL **locks (freezes) at the band edge on first touch** — a path-dependent barrier, not a European condor.

Throughout, **notional** (equivalently, *position capital*) means the LP capital deployed at entry — not the virtual-liquidity notional inside the Uniswap range, which for concentrated positions differs. Surface snapshot: Deribit ETH options, listed expiries, SABR fit dated 2025-06-07.

Indicative IL-protection premiums (model fair value, % of notional), priced off the smile:

Series	10d	30d	60d	90d
V2	0.10%	0.60%	0.86%	1.16%
V3R 50-150	0.43%	2.60%	3.60%	4.66%
V3L 50-150	0.43%	2.35%	2.88%	3.50%
V3R 70-130	0.72%	3.94%	5.17%	6.47%
V3L 70-130	0.72%	2.76%	3.23%	3.92%

These premiums are the price at which an LP converts a variable IL exposure into a **fixed, upfront cost**. Whether fee income covers that cost is concrete (Table 2): at the blended fee APR, 30-day fees are ~1.3% of capital, so a full-range hedge is comfortably covered while tightly concentrated hedges cost more per dollar of capital.

Series	30d hedge	30d fees*	cover (fees/hedge)	breakeven**
V2	0.60%	1.29%	2.1x	14 d
V3R 50-150	2.60%	1.29%	0.5x	61 d
V3L 50-150	2.35%	1.29%	0.5x	55 d
V3R 70-130	3.94%	1.29%	0.3x	92 d
V3L 70-130	2.76%	1.29%	0.5x	64 d

Table 2. Fee-cover and breakeven for the 30-day hedge, as % of position capital. *30d fee income at the blended 15.7% APR (flat, capital-efficiency-agnostic). **Holding days at which cumulative fees equal the premium (premium ÷ fee APR). Full-range (V2) protection is comfortably fee-covered (~2x; breakeven ~14 days); tightly concentrated hedges cost more at a flat APR, but concentration earns a *higher* fee APR (capital efficiency) that scales both the fee numerator and the premium, which could pull the V3 ratios back toward V2's, depending on realised fee density.

In hedge terms: **for the LP** — buy IL protection, keep the fees, retain the residual upside; **for the protection seller** — collect the premium for warehousing the LP's short-gamma risk. The product thesis is that, for full-range and moderately concentrated positions, fee income meaningfully defrays the cost of hedging IL exposure.

3. The rebalancing gamma bleed, quantified

Topaz Blue and Bancor explicitly compared rebalancing to **delta-hedging gamma bleed** — once rebalanced the loss is no longer impermanent — and left the Monte-Carlo analysis to future research; this section supplies it. We calibrate to on-chain realised vol and fee/TVL data, with the path *shape* informed by the fitted options skew: 90-day realised ETH volatility of 41% (Dune), a TVL-weighted blended fee APR of 15.7% across the 5 pools, and the fitted SABR skew. Fig3 reports mean net return versus HODL and the share of paths finishing below HODL, for passive-hold versus rebalance-on-exit.

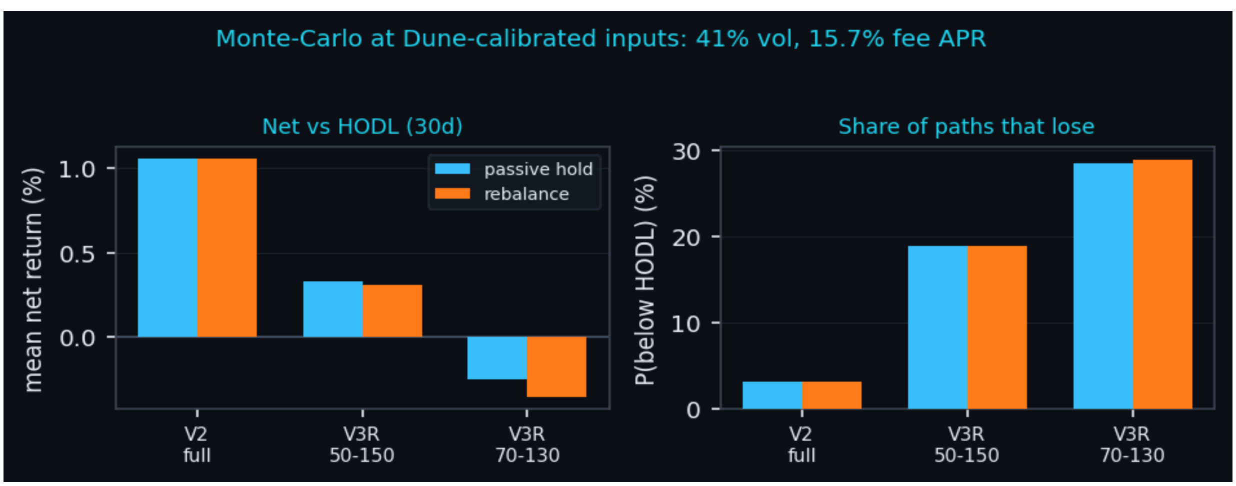

Fig3. Calibrated Monte-Carlo (30-day horizon). Concentration sharply raises the loss probability; the hold-vs-rebalance gap is small at this horizon (see text), and V2 full-range never exits its band, so its two bars are identical by construction.

Model details. Each scenario simulates **40,000** price paths over a **30-day** horizon in **90** time-steps (≈8-hour steps) under the lognormal β=1 SABR SDE ($dF=\alpha F\, dW_1$, $d\alpha=\nu\alpha\, dW_2$, corr ρ) with the fitted ρ and ν and a **drift of zero** (we take no directional view; vol level is the Dune realised figure). Fees accrue at the pool APR while price is **in range**. The rebalance rule, on a band exit, **re-centres** the band at spot, **crystallises** the IL accrued since entry, and pays a **15 bp** cost; V3L instead freezes IL at the edge on first touch. Net return per path = fees + IL − costs; P(loss) is the share of paths with net < 0. Results are stable to the seed at this path count; the live dashboard exposes all of these as adjustable inputs.

At these inputs the wide full-range position is mildly profitable (mean net +1.1%, P(loss) 3%), but a tightly concentrated position turns negative (V3R 70–130 mean net -0.3%, P(loss) 28%), and rebalancing-on-exit makes it worse still — the bleed the original study flagged. The hold-vs-rebalance bars look **nearly identical** because, at a 30-day horizon, two effects roughly cancel: re-centring keeps fees flowing (a *passive* position out of range earns nothing), while each band exit **crystallises** the IL accrued since entry and pays a ~15 bp cost. The bleed is therefore real but second-order here; it **compounds** with horizon, volatility and the number of band exits, so over longer holds or tighter, more-frequently-breached bands the gap widens materially (V2 full-range never exits, so its two bars coincide exactly). The practical implication closes the loop with Sections 1–2: because the loss is a priced, tradeable short-gamma exposure, a fairly-valued IL hedge purchased off the smile can move the LP's expected outcome from negative back to positive, financed by fee income.

Limitations. The measured IL of Section 1 is a clean *subset* (single-cycle, ≥1-day, closed positions) and is therefore a validation of direction rather than a pool-wide total; it is priced at CEX minute prices rather than the pool's own tick price, and, as the original authors stress, IL in USD is not invariant under a change of numeraire when capital is added and withdrawn over time; the single-cycle, closed-position filter may also underweight professional, actively-managed (multi-cycle) LPs, so the sample is clean but behaviourally selected. Separately, because first-touch (V3L) protection is **path-dependent**, its price is **model-dependent even when the terminal smile is matched** — the barrier value is not pinned down by the vanilla smile alone. These limitations constrain the dollar-total interpretation, but not the directional validation.

Appendix A. On-chain methodology (Dune Analytics)

Every input was computed from public Ethereum data on **Dune Analytics**; query design, debugging, and exposition were carried out with **Claude AI (Opus 4.8)**. The position-level IL estimator required 8 iterations (Table A1): each intermediate failure was diagnostic, and the decisive fix was valuing each position at its *exact exit-minute* price.

Table A1. 8-step derivation of the position-level IL estimator.

Step	Estimator	Outcome	Lesson
1	aggregate Mint - Burn, valued at close	mixed (1 gain)	net flow x price is not IL
2	per-position match by owner + tick range	3/5 negative	owner = NFT manager collapses positions
3	same match, valued at exit-day price	4/5 positive	owner+tick matching is broken - need token-ID
4	NFPM token-ID, exit-day price	3/5 negative	right source; multi-cycle positions leak
5	single deposit->withdraw cycle only	4/5 negative	JIT / flash pool still positive
6	+ hold >= 1 day (exclude JIT)	4/5 negative	not only JIT - deeper cause remained
7	all-time deposit capture	unchanged	window truncation ruled out
8	value at exact exit-MINUTE price	5/5 negative	IL identity holds only at the exit price

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